Interactive Data Visualization in Qualitative Research

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Qualitative researchers analyze data taken from a variety of sources towards the purpose of understanding how and why people do what they do. These researchers almost always gather this data in the form of text: field notes, interview transcripts, transcribed video and audio tapes, archival records, etc., and then analyze it through various coding and categorizing procedures (LeCompte, Millroy, & Preissle, 1992). Until the recent development of computer programs to support this process (see Fielding & Lee, 1991), qualitative data analysis had always been a manual operation involving the cutting, pasting, sorting and reviewing of hundreds of text fragments a few at a time. Even with this software support, qualitative researchers still suffer from acute difficulties in visualizing and representing relationships among data and so in distinguishing and communicating which are the most meaningful. Fortunately, this too is being changed by the new software.

Figure 1 shows a "data map," an interactive graphic created with a hypertext editing program called *EntryWay* (Horney, 1992). Beyond the basic coding and sorting operations, programs such as *EntryWay* also include functions for creating graphic visualizations of data and data relationships. Figure 1 is an example of such a map. It shows categorical relationships among 100 messages (part of a university policy discussion) posted on an electronic bulletin board and was made as part of an analysis of different styles of electronic communication. *EntryWay* maps such as this are created by first importing the messages into *EntryWay*, segmenting them so that each message was held on a separate hypertext node, and by then coding the messages by placing them into categories, or "threads" as they are called in *EntryWay*, to denote characteristics such as authorship and message content.

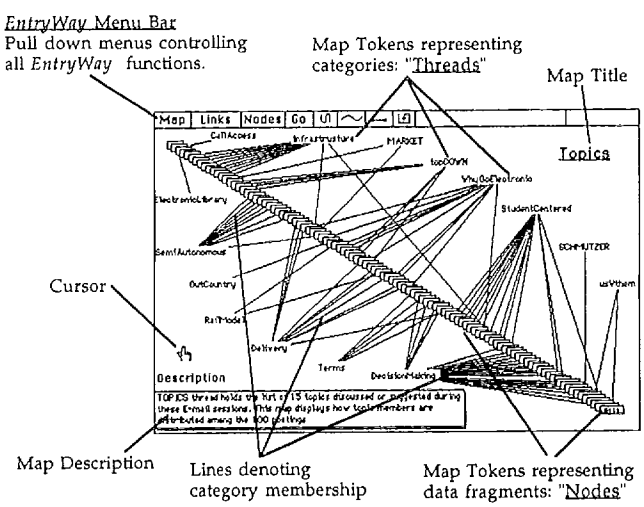

Figure 1. Map of the distribution of topics among e-mail messages

Once the coding process has begun, maps can be started to give researchers an ongoing visual representation of the categorical relationships resulting from the coding. On a map, nodes are represented by "tokens." In the map in figure 1, the messages are represented by the tokens drawn down the diagonal and message topic categories by the tokens around the map edges. The lines drawn from the category tokens to the message tokens show categorical membership. Researchers make maps through an iterative process of placing tokens on the map using several retrieval functions, (re)arranging them in meaningful patterns and by then using *EntryWay* to automatically draw the inclusion lines.

Maps can also show semantic relationships among nodes. The comment map shown in Figure 2 is based on binary links rather than the categorical links used in Figure 1. In this case, these "replied to" links show how the messages were entered as replies to one another. For instance, the cluster of nodes in the upper left corner are the nodes of the "Infrastructure" category. This topic was initiated in message G14. Message G15 was entered as a reply to G14, and was in turn commented upon by message G16. There were two replies to G16, and so on. Thus, this map shows the message reply structure of each conversational topic. Link-based maps are created in the same way that thread-based maps are, and both threads and links can be represented on the same map.

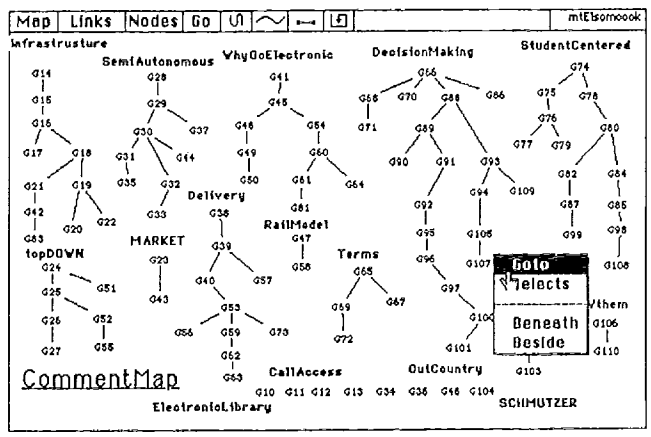

Figure 2. Map of reply relationships among e-mail messages

Examples of data visualizations in EntryWay

To date, seven different educational researchers have made data maps using *EntryWay*. From their experiences four specific uses for data maps have emerged: (a) as navigational aids; (b) as visualizations prompting further investigation; (c) as formal representations of relationships; and (d) as cognitive supports during coding.

Figure 2 captures a map in use as a navigational aid. A reader is shown selecting the "GOTO" option from a menu which appears whenever a map token is clicked by the cursor. In this way, researchers can move about their data, finding and reading relevant selections.

As visualizations, data maps can bring into view subtle relationships that might otherwise go unnoticed. For instance, Figure 3 suggests a contradiction in teachers' ideas about their adminis-

trators. This map shows teachers (represented by tokens down the diagonal) making many negative comments about school administrators (categories "Impossible," "DoNothing" and "TreatDiff"), but yet also report "Trust(ing)" administrators. This contradiction in teacher attitudes was overlooked until made explicit by the map.

Similarly, after looking at Figure 4, another researcher realized that the "ASwar" and "ASrevie" categories were so similar, as actually used during coding, that they should be merged. Maps can also point out areas where data analysis should be reconceptualized. For example, after seeing in maps such as Figure 5, how whole families of coding categories were being under utilized, the researcher realized that her basic coding strategy was not working and so instituted a new set of categories. In these ways data maps lead researchers into deeper and alternative investigations of their data.

More formal maps can serve as representations of particular features of their data that researchers wish to communicate to others. For instance, Figures 6 and 7 demonstrate the differences between two students' reaction to a story. Clint was frustrated by the assignment and repeatedly stated during an interview that he was "Done" with the reading assignment, even though he responded "Don't Know" to most questions, and frequently asked if the interview was "Finished." Sue, on the other hand, was able to supply a wide variety information about what she had read.

The last use for data maps is encapsulated in the aphorism "Making a map is better than having a map." It appears that map making gives researchers the opportunity to quietly consider their data as they add and remove nodes, move tokens from place to place, draw and erase link and inclusion lines, and by (literally) untangling relationships during the creation of coherent displays. The result of this contemplation may be new insights into the data and its implications.

Making diagrams of relationships among data is not new, nor are programs like *EntryWay* in any way optimal. However, the introduction of interactive graphics gives new tools to researchers in a field that has been overwhelmingly dominated and burdened by the limitations of text. These, together with computerized data management, and new techniques of multimedia data collection and analysis (Goldman-Segall, 1990), should invigorate qualitative researchers and bring their work into view in ways never before anticipated.

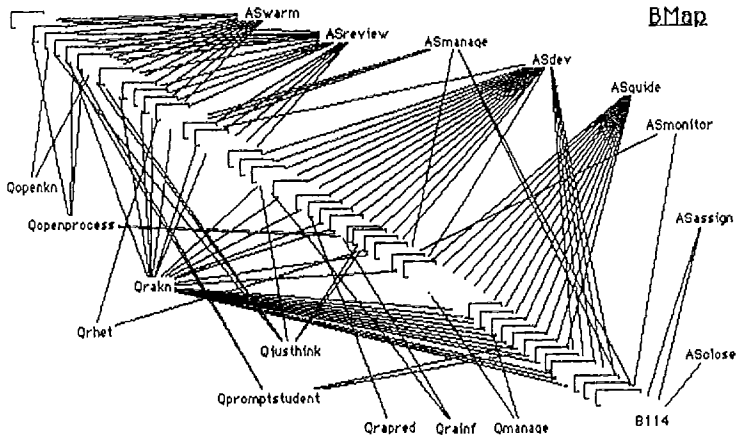

Figure 4. Coding of classroom activities and teacher questions during a math lesson

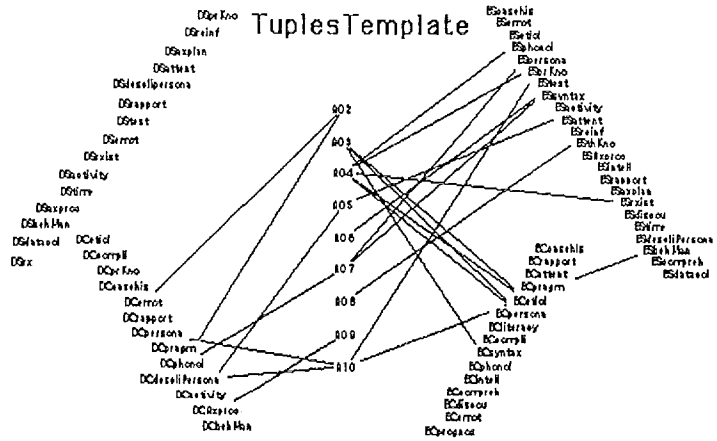

Figure 5. Coding of student A's evaluation of her first clinical experience

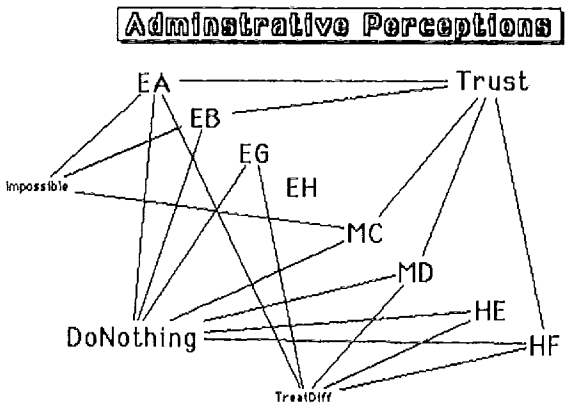

Figure 3. Coding of teacher perceptions of administrators

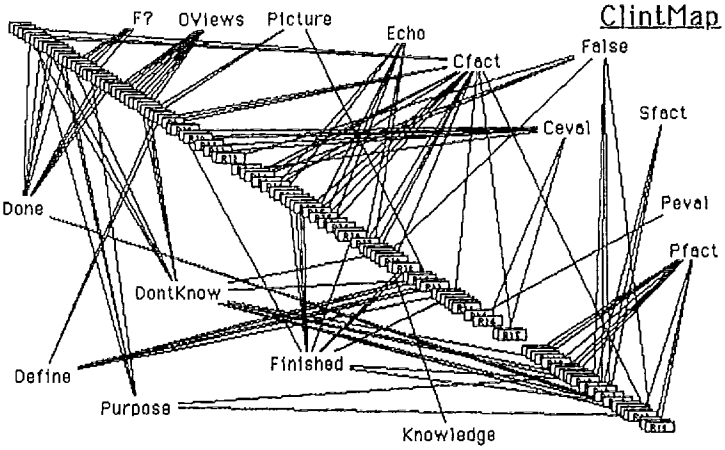

Figure 6. Coding of Clint's retelling of a story

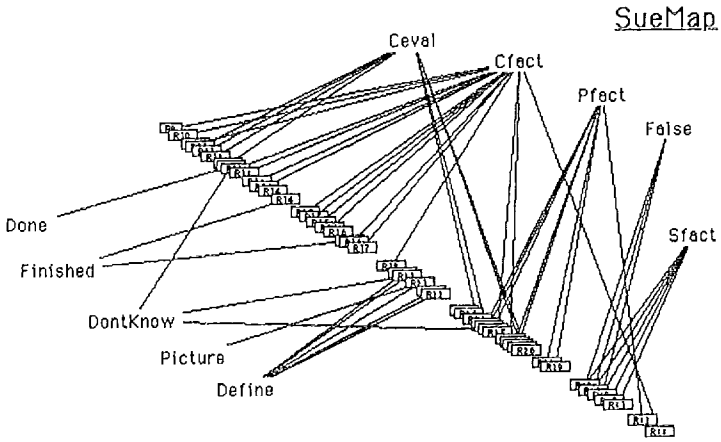

Figure 7. Coding of Sue's retelling of a story

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Research–Guided Design of Multimedia Research Tools

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Since this issue of the newsletter features both multimedia designers and research-oriented users, I thought it would be interesting to merge the two viewpoints and describe my approach to the design of some of the multimedia software I use in my research. The main point of this article is to describe how the findings of prior research can be used to guide the design of software which itself is to be used for additional research. The beginning of the discussion will center around the development of a C program called "VideoGraph." The name furnishes a hint as to what the program does—it provides introductory physics students with a tool for graphically analyzing video sequences. A second package was designed to provide a multimedia editing environment for use by middle school students. Although these two projects sound very different, the software for both shares the same purpose—it gives students multimedia tools to help them learn science topics while minimizing the distraction of the software itself. The research investigated the educational impact of the software.

VideoGraph

The main task of the VideoGraph package is to replay a video of one or more moving objects while synchronously generating a graph of position or velocity. Many studies have shown that students have a great deal of difficulty making the cognitive leap from the concrete world of motion events to their abstract mathematical—in this case, graphical—representations (Barclay, 1986; McDermott, Rosenquist, & Zee, 1987; Mokros & Tinker, 1987; van Zee & McDermott, 1987; Brasell & Rowe, 1989; Beichner, 1993). Students tend to view graphs as a sort of photograph of the scene. For example, when asked to sketch a velocity versus time graph for an object rolling down a hill, across a flat area and then back up a hill, the resulting drawing will often reproduce the hills and valley traversed by the object. When shown the actual kinematics graphs, students are not able to interpret their meaning.

Besides the problems students have in understanding kinematics graphs, they also have some basic difficulties with other aspects of Newtonian mechanics (Trowbridge & McDermott, 1980; Trowbridge & McDermott, 1981; Halloun & Hestenes, 1985; Hestenes, Wells, & Swackhamer, 1992). For example, students believe (even after traditional instruction) that there must be a force on an object if it is moving. The classic test for this is to toss a coin into the air and ask students to describe any force which might be acting on it during its flight. When interviewed, the common response is to describe a "force from the throw" that decreases as the coin rises. The peak of the projectile's motion takes on an inappropriate level of importance to these students when in fact the acceleration of the coin remains constant throughout the throw. (A graph of the velocity is a straight, downward sloping line which crosses the time axis at the instant the coin reaches the top of its arc.)

Rather than turn this article into a physics lesson, let us briefly turn our attention to some of the psychology behind the user interface of VideoGraph. Keep in mind that the overall plan for the software is to address the problems research has uncovered about students trying to make the mental links between concrete events and the related graphs. This is directly reflected by the decision to have two main windows, one for the event and the other for a graph.

The video window utilizes QuickTime™ to present a movie of the motion. It is placed directly beside the graph window. Because short term memory is limited in what it can hold and how long its contents can be maintained (Hulse, Egeth, & Deese, 1980) , I decided to keep the motion event directly linked to an adjacent graphical representation. Students can easily go back and forth between the two "worlds." These two displays are synchronized so